

1 Problem

- Every fully fine-tuned task duplicates all pretrained weights and optimizer states; GPT-3 175B makes this prohibitive.
- Adapters add sequential layers and inference latency, while prompt tuning consumes context and can be hard to optimize.
- Hypothesis: downstream adaptation changes occupy a much lower intrinsic rank than the original dense weights.

2 Method

FACTORIZE THE UPDATE

- $W = W_0 + \Delta W$ ;  $\Delta W = B A$ ;  $h = W_0 x + B A x$
- Freeze  $W_0$ ; train  $A$  in  $R^{(r \times k)}$  and  $B$  in  $R^{(d \times r)}$ , with  $r$  much smaller than  $d$  or  $k$ .
- Initialize  $A$  with Gaussian noise and  $B$  at zero, then scale the branch by  $\alpha/r$  so training starts exactly at the base model.

3 Where to adapt

ATTENTION PROJECTIONS

- LoRA can target any dense layer; experiments focus on Transformer query, key, value and output projections.
- With a fixed parameter budget, adapting both  $W_q$  and  $W_v$  at small rank usually beats allocating a larger rank to one matrix.
- Ranks as low as 1-4 can be competitive on GPT-3 tasks even when the hidden dimension is 12,288.

4 Systems benefits

- Store one frozen foundation model plus small task-specific LoRA modules and switch tasks by swapping A/B factors.
- Merge BA into  $W_0$  at deployment, adding no inference latency versus the original model.
- For GPT-3 175B, training memory falls from about 1.2 TB to 350 GB and checkpoints from 350 GB to 35 MB in the reported setting.

5 Evidence

MATCH OR BEAT FINE-TUNING

- LoRA matches or improves full fine-tuning on RoBERTa, DeBERTa, GPT-2 and GPT-3 with far fewer trainables.
- GPT-3 uses only 4.7M LoRA parameters for strong WikiSQL, MultiNLI and SAMSum results versus 175B full parameters.
- Compared with GPT-3 Adam fine-tuning: about 10,000x fewer trainables, 3x lower GPU memory and higher training throughput.

6 Meaning & limits

ADAPTATION IS LOW-RANK

- Learned updates amplify task-specific directions not emphasized in the pretrained weight's top singular subspace.
- Optimal rank and target matrices remain task- and model-dependent; scaling relations were left open.
- LoRA reduces trainable state but does not quantize the frozen base—QLoRA later attacks that dominant memory cost.

LORA

LoRA

Low-Rank Adaptation of Large Language Models | 2021

Freeze the pretrained model and learn each task's weight change through tiny low-rank factors, preserving model quality while sharply reducing optimizer memory, checkpoint size and deployment overhead.

PEFT

low-rank update

zero inference overhead